PROBABILITY & STATISTICS PROJECT

Unemployment Analysis in Pakistan Using Regression

Submitted to: Ma'am Warda

Abdullah 22i-2346

Wasif Mehmood 24i-0699

Hammad Rasheed 24i-0703

Section F

1.Introduction:

Unemployment has long been one of the most stubborn economic problems facing developing countries, and Pakistan is no exception. Over the decades, the country has grappled with a combination of rapid population growth, large-scale urban migration, and major technological shifts — all of which have left a visible mark on the labour market.

This study takes a closer look at what drives unemployment in Pakistan, examining data that stretches all the way from 1960 to 2024. Five key factors are brought into focus: Labor Force Participation, Urbanization, Internet Penetration, Energy Consumption, and Foreign Direct Investment. Together, these variables paint a comprehensive picture of the economic forces at play.

To make sense of the data, two different modelling approaches are used — Multiple Linear Regression (MLR) for its interpretability, and Random Forest for its predictive strength. Beyond identifying relationships, the study also compares how well each model performs, using standard error metrics like MSE, RMSE, and MAE.

2. Literature Review

Economic theory has long recognized that unemployment is shaped by forces on both the supply and demand sides of the labour market.

When more people join the workforce — reflected in rising Labor Force Participation — unemployment can climb if the economy isn't creating jobs fast enough to absorb them. Urbanization tells a similar story: while cities do generate employment opportunities, they also attract waves of rural migrants, intensifying competition for available jobs.

Internet penetration adds another layer of complexity. On one hand, digital connectivity opens new forms of employment; on the other, it disrupts traditional industries and displaces workers who lack the skills to adapt. Energy consumption, meanwhile, has long been treated as a reliable indicator of industrial activity, which in turn drives job creation. And while FDI inflows are generally expected to help reduce unemployment, their actual impact in Pakistan has depended heavily on which sectors attract the investment and how large those flows are.

Taken together, research on Pakistan consistently points to one conclusion: unemployment here is not a simple, linear problem — it's deeply structural and shaped by long-term economic transformations.

3. Description of Variables and Data

3.1 Data Source and Processing

All data was pulled from the World Development Indicators and aligned by country and year. Once compiled, the dataset was sorted chronologically and cleaned using forward fill (LOCF) and backward fill (BOCF) to handle missing values. After imputation, any remaining gaps were removed, leaving a clean, consistent time-series dataset ready for analysis.

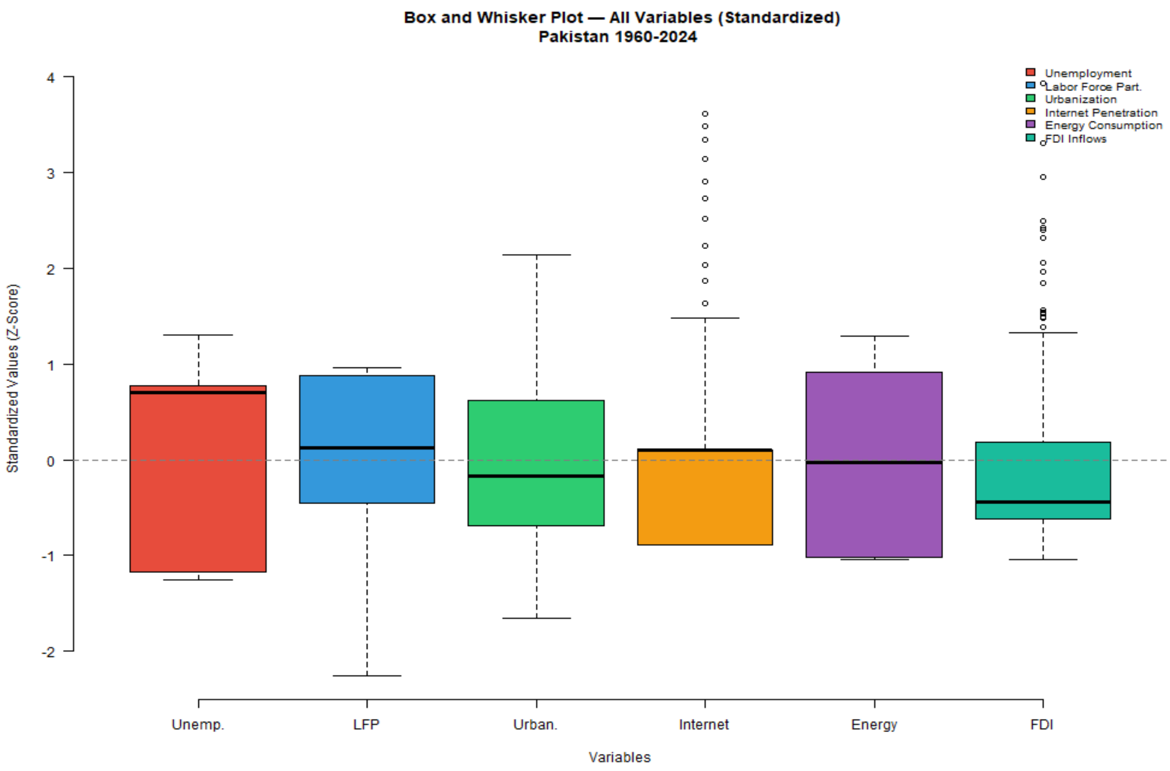
3.2 What Each Variable Represents

Variable	Role	Description
Unemployment	Dependent	% of labour force unemployed
Labour Force Participation	Independent	% of working-age population
Urbanization	Independent	% urban population
Internet Penetration	Independent	% internet users
Energy Consumption	Independent	kg per capita
FDI Inflows	Independent	% of GDP

4. Box and Whisker Plot

Below are the Box and Whisker Plot as each variable as different units, so we have decided to use standardized units. The standardized boxplots reveal that unemployment and urbanization behave relatively predictably, while labour force participation shows wider variation due to shifting workforce dynamics. Internet penetration and FDI inflows stand out the most, with notable outliers reflecting rapid digital growth and sudden investment spikes tied to specific economic events. Energy consumption, while variable, remains within a normal range.

The main thing that graph highlights that Pakistan economy has faced real structural shocks over this period, and those shocks are visible in the data.



5. Scatter Plot Analysis

The scatter plots help reveal the nature of each variable's relationship with unemployment:

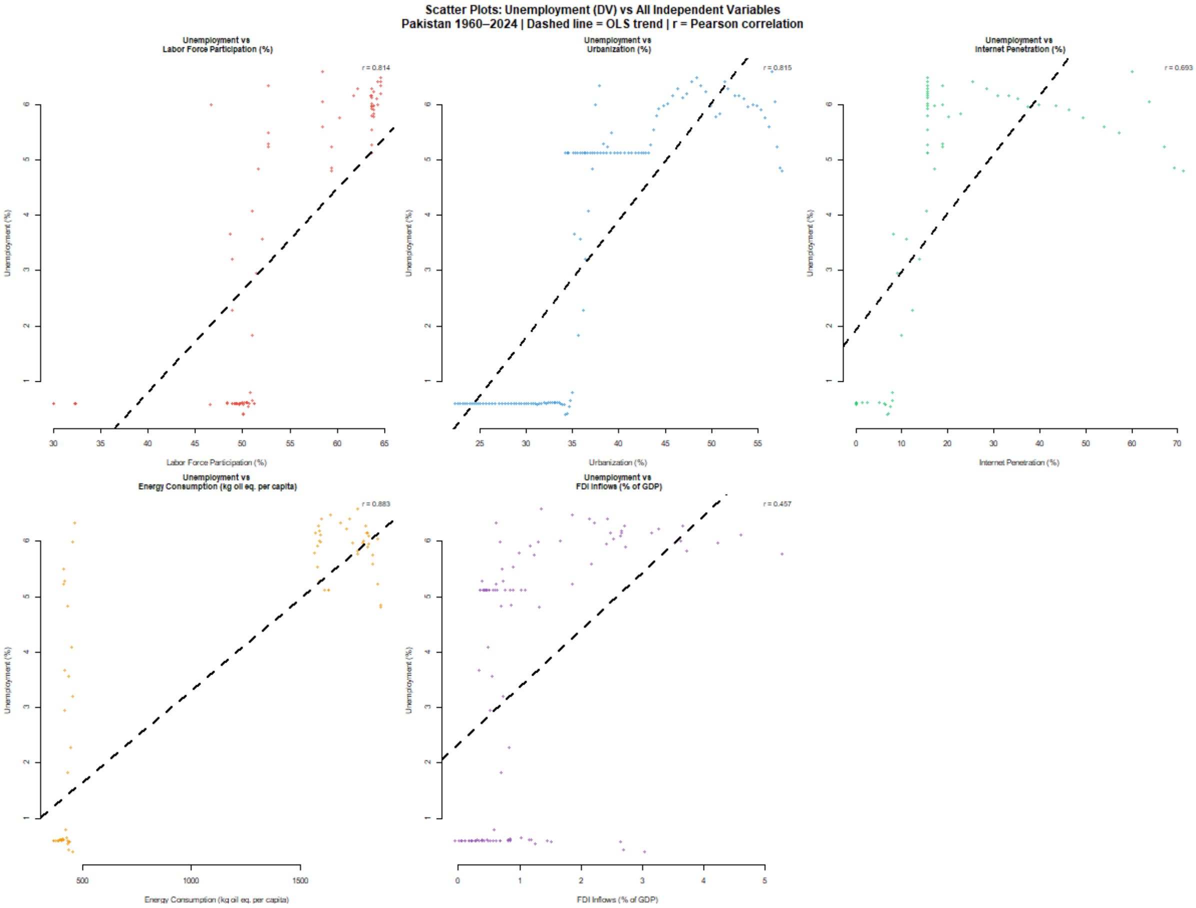
Labor Force Participation shows a strong positive relationship means when more people enter the labour market, unemployment tends to rise, suggesting that job creation hasn't kept up with labour supply growth.

Urbanization follows a similar pattern. As more of the population moves into cities, unemployment goes up, likely because urban labour markets become increasingly crowded.

Internet Penetration shows a moderate relationship. This makes sense as digital adoption creates new jobs, but it also displaces workers in traditional sectors, and the net effect is neither strongly positive nor negative.

Energy Consumption has a strong association with unemployment, reinforcing its role as a proxy for economic activity. That said, the direction of the relationship reflects long-run structural changes rather than a simple cause-and-effect story.

FDI Inflows show a weak and inconsistent relationship with unemployment — suggesting that, at least in Pakistan's case, foreign investment alone hasn't been a reliable engine of job creation



6. How the Models Works

6.1 Multiple Linear Regression:

The regression model is built around a straightforward formula:

$$Unemployment \ = \beta_0 + \ \beta_1 \ + \ \beta_2 + \beta_3 + \beta_4 + \beta_5 + \varepsilon$$

Where:

β_1 = LFP

$\beta_2 \ = \ Urbanization$

$\beta_3 \ = \ Internet$

$\beta_4 \ = \ Energy$

$\beta_5 \ = \ FDI$

Each coefficient tells us how much unemployment is expected to change for a one-unit increase in the corresponding variable. Positive coefficients point to variables that push unemployment higher; negative ones suggest the opposite.

6.2 Random Forest

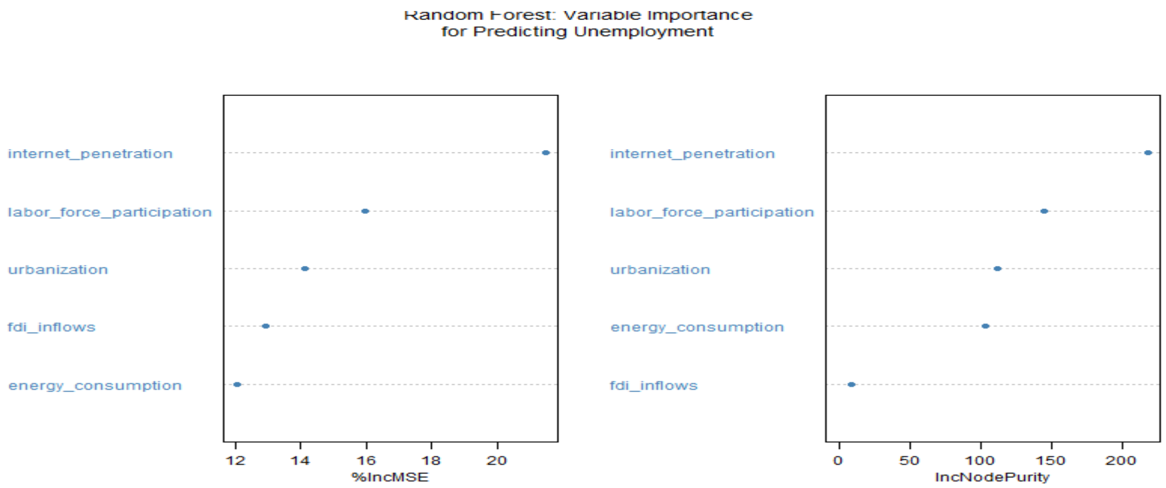
Random Forest works quite differently. Rather than fitting a single equation to the data, it builds many decision trees and averages their predictions. This approach naturally picks up on non-linear relationships, interaction effects, and complex patterns that a standard regression would miss entirely — and it does so without requiring the same strict assumptions.

7. Results

Which variable matters the most?

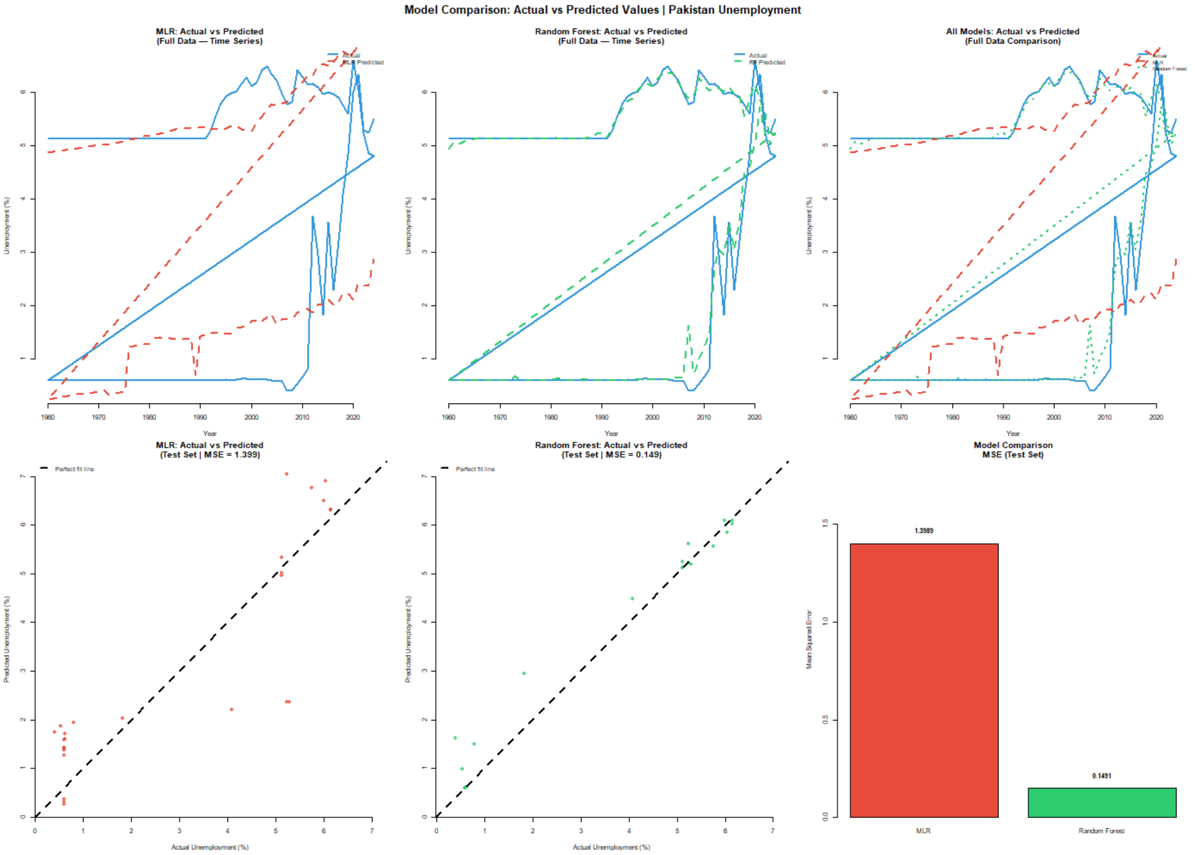
From the figure below using the Random Forest model, the three most influential drivers of unemployment in Pakistan are:

1. Energy Consumption
2. Internet Penetration
3. Labor Force Participation



This suggests that economic activity levels and technological change are more central to unemployment dynamics than investment flows. FDI, at the other end of the scale, has the least predictive importance.

7.2 How Well Do the Models Perform?



7.3 What the Comparison Reveals

The regression model does a reasonable job of capturing the general direction of unemployment trends, but it struggles when it comes to fluctuations. Its predictions tend to be smooth and slow to react to sudden changes.

The Random Forest model, by contrast, tracks actual unemployment much more closely by picking up both the broader trends and the shorter-term movements. When you plot predicted versus actual values, the Random Forest predictions cluster tightly around the ideal 45-degree line, which is exactly what you want to see.

The lower MSE confirms what the charts suggest: Random Forest is simply a better-fitting model for this data. And the reason comes down to the nature of unemployment itself — it's driven by complex, interacting forces that a linear model can't fully capture, but a tree-based ensemble can.

8. Conclusion

This study set out to understand what drives unemployment in Pakistan, and the findings offer a clear picture.

On the supply side, growth in Labor Force Participation and Urbanization puts upward pressure on unemployment — more workers entering the market than the economy can readily absorb. Internet penetration reflects the structural transformation underway in Pakistan's economy, while Energy Consumption emerges as one of the strongest indicators of whether the economy is generating enough activity to create jobs. FDI, despite its theoretical promise, has had limited and inconsistent effects on employment.

As for the modelling side, Random Forest clearly outperforms Multiple Linear Regression — and that outcome itself is informative. It tells us that unemployment in Pakistan is not a linear phenomenon. It's shaped by non-linear dynamics and variable interactions that require more flexible tools to properly model.

The broader takeaway is that reducing unemployment in Pakistan will require more than any single policy lever. Economic growth, technological development, and labour market reform need to work in tandem — because in a complex economy, they all pull on the same thread.

References:

World Bank (World Development Indicators)

https://databank.worldbank.org/source/world-development-indicators